COMP3026E: Distributed Systems

Assignment 1

Deadline: 20:00 (PM), Apr. 17th Friday, 2026

Submission: Name your (scanned/e-) answer sheet by `A1-{Student ID}-{Student Name}.pdf` excluding both curly brackets. Send it to teaching assistant Mr. Chao WANG (QQ: 3444813826) in the course QQ group.

Part 1: Short Answer Questions (3 points each)

1. What is the primary objective of a Remote Procedure Call (RPC)?
2. What are the specific roles of the client stub and the server stub in the RPC lifecycle?
3. Why is Coordinated Universal Time (UTC) via GPS not a universal solution for all networked computers?
4. In Cristian's Algorithm, how does a client adjust its clock after receiving a response from a time server?
5. What are the primary reasons a distributed system requires global snapshots?
6. According to the Chandy-Lamport algorithm, what must a node do immediately after recording its local state?
7. What three components constitute a "write" bundle in the Bayou system?
8. How does the "Primary Commit" scheme in Bayou establish a total order for writes?
9. Why is the standard $\mathrm{mod}(N)$ approach insufficient for dynamic P2P systems?
10. What is the fundamental assumption required for State Machine Replication (RSM) to function?

Part 2: Long-Form Analysis

1. Peer-to-peer file-sharing with Chord. (10 points each)

   Task A: Explain how Finger Tables improve lookup efficiency compared to a simple successor-based ring. Specifically, reference the number of messages required for a lookup in a system with N nodes.

   Task B: Describe the stabilization process that occurs when a new node joins the Chord network. How do nodes maintain correct successor and predecessor pointers?

2. Logical Time and Causality. (10 points each)

   Task A: Explain the primary limitation of Lamport Clocks regarding causality. Specifically, if you are given two events a and b where the timestamp $C(a) < C(b)$, why can you not conclude that a causally preceded b ($a \to b$)?

   Task B: Describe how Vector Clocks address this limitation. Explain the structure of a Vector Clock timestamp and how it allows a system to determine if two events are concurrent ($a \parallel b$) rather than causally linked.

Part 3: Comparative Design Questions

1. Compare and contrast the Bayou approach to consistency with the Primary-Backup Replication model. (6 points each)
   - Which system is better suited for "disconnected operation" (e.g., a mobile device with intermittent Wi-Fi)?
   - How does each system handle the "Source of Truth"?
   - Explain the concept and implementation of "Read Your Writes" in the context of Bayou's session guarantees.

2. When designing a Remote Procedure Call (RPC) system, failures can lead to subtle bugs. (6 points each)
   - Compare "At-least-once" and "At-most-once" delivery semantics. If you are designing a system to process credit card transactions, which of these is safer, and why?
   - Describe the additional mechanisms required (such as fault tolerance and duplicate filtering, maybe more) to achieve "Exactly-once" semantics in a distributed environment.